Project code: chat-deployment

The code for this example project is in the ch09/chat-deployment folder. You can use that example by running this command in the root folder of the project repository:

```
$ npm run dev -w ch09/chat-deployment
```

You'll need the Google API key, the existing Upstash keys, and also the Clerk.js authentication keys to access their respective REST APIs. These are configured as environment variables:

```
// .env
GEMINI_API_KEY=<GEMINI_AI_SECRET_KEY>
UPSTASH_REDIS_REST_URL=<UPSTASH_REDIS_REST_URL>
UPSTASH_REDIS_REST_TOKEN=<UPSTASH_REDIS_REST_TOKEN>
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=<NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY>
CLERK_SECRET_KEY=<CLERK_SECRET_KEY>
```

Instructions on how to set up and create secret keys for Google, obtain the necessary credentials for Google AI and Upstash Redis, as well as how to retrieve your Clerk API keys are provided in the appendix.

Since we will be deploying to Vercel, we can check many items from the list regarding security and quality of service items, including

- *Encrypting sensitive data at rest and in transit*—Vercel deployments are served under HTTPS, and we use external providers for authentication, like Clerk, which securely encrypts user data at rest.
- *Monitoring and logging*—Vercel offers detailed monitoring features even in its free plan, allowing us to keep track of application performance and problems.
- *Implementing CI/CD pipelines*—Vercel integrates seamlessly with Git, enabling automated CI/CD pipelines through its deployment and build tools.
- *API keys*—Vercel provides a secure way to manage and inject environment variables into your applications, ensuring sensitive information is protected.
- *Cost management*—We will limit interactions with the model to 10 messages per day per user. Additionally, access will be restricted to select users for testing purposes only.

Before we deploy to Vercel, you need to first create an account and enter your business or personal details. You can sign up by visiting the Vercel website at https://vercel.com/ and choosing your preferred plan.

To create an account, simply enter your email address to receive a six-digit one-time password, or you can sign up using a supported Git provider for easier integration. After registration, you'll need to verify your email address to complete the setup.